

The **multinotes** package*

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A package for typesetting multilingual lecture notes and exercise sheets.

1 Introduction

This package provides tools for typesetting multilingual lecture notes in various formats, and adopts some macros and environments from the **exam** document class for producing exercise sheets and solutions.

The package repository is at

<https://github.com/dimbyd/multinotes/>

2 Package options

The following package options control the visibility of different elements.

blanks	Remove contents of <code>\blank</code> cmds and blankbox envs.
noproofs	Exclude <code>\prf</code> commands and proof environments.
blankproofs	Replace <code>\prf</code> commands and proof environments by blanks.
noanswers	Exclude <code>\ans</code> commands and answer environments.
blankanswers	Replace <code>\ans</code> commands and answer environments by blanks.
nosketchboxes	Exclude sketch boxes.
student	Equivalent to choosing blanks , blankproofs and noanswers .
compact	Equivalent to choosing noproofs , noanswers and nosketchboxes .

3 Languages

A language is specified by its **babel** name.

`\definelanguage` Define the declaration name, macro name and environment name.
`\setlanguagestyle` Set language-specific style options (fonts, colours, etc.).

`\includelanguage` Include language in document.

`multicols` (*env.*) Environment for parallel typesetting (one column for each language).

`\multicolson` Switch parallel mode on.

`\multicolsoff` Switch parallel mode off.

`\multicolspan` Typeset content across all columns.

`\multicolsync` Synchronize columns.

4 Lecture notes

`blankbox` (*env.*) Environment for displayed blank box (responds to `blanks` option).

`\blank` Command for inline blank box (responds to `blanks` option).

`\prf` Command for inline proof box (responds to `blankproofs` and `noproofs`).

`proof` (*env.*) Environment for displayed proof box (responds to `blankproofs` and `noproofs`).

`\ans` Command for inline answer box (responds to `blankanswers` and `noanswers`).

`answer` (*env.*) Environment for displayed answer box (responds to `blankanswers` and `noanswers`).

`\filin` Command for inline fill-the-blank box (responds to `blankanswers`).

`\textcolour` Set default font colour.

`\backgroundcolour` Set background colour.

`\showcolour` Set font colour for blank boxes.

`\proofcolour` Set font colour for proof boxes.

`\setanswercolour` Set font colour for answer boxes.

`\framedblankboxes` Choose framed blank boxes (inline and displayed).

`\framedproofs` Choose framed proof boxes (inline and displayed).

`\framedanswers` Choose framed answer boxes (inline and displayed).

`\setboxrule` Set width of the frame border (default is zero).

`\setstretchfactor` Set stretch factor for blank boxes (to accomodate handwriting).

`\setimagestretchfactor` Set stretch factor for blank images (to accomodate hand-drawing).

`\blankson` Override global `blanks` option.

`\blanksoff` Override global `blanks` option.

`\blankproofson` Override global `blankproofs` option.

`\blankproofsoff` Override global `blankproofs` option.

`\noproofsoff` Override global `noproofs` option.

`\noproofson` Override global `noproofs` option.

`\blankanswersoff` Override global `blankanswers` option.

`\blankanswerson` Override global `blankanswers` option.

`\noanswersoff` Override global `noanswers` option.

`\noanswerson` Override global `noanswers` option.

*This document corresponds to `multinotes` v1.0, dated 2025/12/31.

5 Exercises

Inspired by `exam.cls`, responds to `blankanswers` and `noanswers` options.

`questions` (*env.*) List environment for questions.

`parts` (*env.*) List environment for parts of a question.

`subparts` (*env.*) List environment for subparts of a part.

`\question` Item type for the `questions` environment.

`\part` Item type for the `parts` environment.

`\subpart` Item type for the `subparts` environment.

`choices` (*env.*) List environment for multiple choice options (choose one).

`checkboxes` (*env.*) List environment for multiple answer options (choose zero or more).

`\choice` Item type for incorrect choices within `choices` and `checkboxes` environments.

`\correctchoice` Item type for correct choices within `choices` and `checkboxes` environments.

6 Implementation

Load packages

```
1 \RequirePackage{amsmath,amsfonts,amssymb,amsthm}
2 \RequirePackage{babel}
3 \RequirePackage{caption}
4 \RequirePackage{comment}
5 \RequirePackage{etoolbox}
6 \RequirePackage{float}
7 \RequirePackage{geometry}
8 \RequirePackage{graphicx}
9 \RequirePackage{kvoptions}
10 \RequirePackage{paracol}
11 \RequirePackage{setspace}
12 \RequirePackage{tcolorbox}
13 \RequirePackage{tikz}
14 \RequirePackage{titling}
15 \RequirePackage{translations}
16 \RequirePackage{trimspaces}
17 \RequirePackage{xifthen}
18 \RequirePackage{xstring}
19 \RequirePackage{xspace}
20 \RequirePackage{xparse}
21 \tcuselibrary{breakable}
22 \tcuselibrary{skins}
```

6.1 Options

```
23 \newif\ifmultinotes@blanks
24 \newif\ifmultinotes@noproofs
25 \newif\ifmultinotes@blankproofs
26 \newif\ifmultinotes@noanswers
```

```

27 \newif\ifmultinotes@blankanswers
28 \newif\ifmultinotes@nosketchboxes
29 \DeclareOption{blanks}{\multinotes@blankstrue}
30 \DeclareOption{noproofs}{\multinotes@noproofstrue}
31 \DeclareOption{blankproofs}{\multinotes@blankproofstrue}
32 \DeclareOption{noanswers}{\multinotes@noanswerstrue}
33 \DeclareOption{blankanswers}{\multinotes@blankanswerstrue}
34 \DeclareOption{nosketchboxes}{\multinotes@nosketchboxestru}
35 \DeclareOption{student}{
36     \multinotes@blankstrue
37     \multinotes@blankproofstrue
38     \multinotes@noanswerstrue
39 }
40 \DeclareOption{compact}{
41     \multinotes@nosketchboxestru}
42     \multinotes@noproofstrue
43     \multinotes@noanswerstrue
44 }
45 \DeclareOption*{\PackageWarning{multinotes}{Unknown option ‘\CurrentOption’}}
46 \ProcessOptions*
47 \relax

```

6.2 Language tools

Languages are referenced by their **babel** names.

6.2.1 Define language

Usage example: `\definelanguage{welsh}{cy}{cym}{cymraeg}`

This will define

- three internal commands: `\@welshdec`, `\@welshcmd` and `\@welshenv`,
- a declaration `\cy` which maps to `\@welshdec`,
- a command `\cym{...}` which maps to `\@welshcmd{...}`,
- an environment ‘`cymraeg`’ which maps to environment ‘`@welshenv`’.
- an environment ‘`welsh`’ which also maps to environment ‘`@welshenv`’.

Important. Language environments are implemented as *comment* environments, which allows us to discard their contents when required. The *comment* package documentation states that

“... all text included between `\begin{comment}` and `\end{comment}` is discarded. The opening and closing commands should appear on a line of their own. No starting spaces, nothing after it.”

Failure to ensure this can produce some very confusing errors! This also applies to the **proof** and **answer** environments defined below.

```

48 \newcommand{\multinotes@definelanguage}[4]{

```

```

49 \ifcsname multinotes@#1dec\endcsname\else
50   \expandafter\def\csname multinotes@#1dec\endcsname{\multinotes@commonstyle\multinotes@ren
51   \expandafter\def\csname #2\endcsname{\csname multinotes@#1dec\endcsname}
52   \expandafter\def\csname multinotes@#1cmd\endcsname##1{
53   \expandafter\def\csname #3\endcsname{\csname multinotes@#1cmd\endcsname}
54   \expandafter\def\csname multinotes@#1envname\endcsname{#4}
55   \specialcomment{#4}{}{}
56   \excludecomment{#4}
57   \expandafter\def\csname multinotes@#1style\endcsname{}
58 \fi
59 }
60 \newcommand{\definelanguage}[1]{\multinotes@definelanguage{#1}}

```

6.2.2 Include language

Usage example: `\includelanguage{welsh}`

When a language is included it is assigned a sequential number, which specifies its column number in *paracol* environments. The `\switchcolumn` command is only defined within *paracol* environments, and therefore provides a criterion to decide whether column switching is required.

```

61 \newcommand{\multinotes@mainlanguage}{UKenglish}
62 \newcommand{\multinotes@selectlanguage}[1]{\ifpackageloaded{babel}{\selectlanguage{#1}}{}}
63 \AtBeginDocument{\multinotes@selectlanguage{\multinotes@mainlanguage}}
64
65 \newcounter{multinotes@numlangs}
66
67 \newcommand{\multinotes@activatelanguage}[1]{
68   \babelprovide[import]{#1}
69   \expandafter\newcounter{multinotes@#1number}
70   \expandafter\setcounter{multinotes@#1number}{\value{multinotes@numlangs}}
71   \stepcounter{multinotes@numlangs}
72   \ifnum\value{multinotes@numlangs}=1\relax
73     \renewcommand{\multinotes@mainlanguage}{#1}
74   \fi
75   \expandafter\def\csname multinotes@#1dec\endcsname{%
76     \ifundefined{switchcolumn}{}{\switchcolumn[\value{multinotes@#1number}]}%
77     \csname multinotes@#1style\endcsname%
78     \multinotes@selectlanguage{#1}%
79   }%
80   \expandafter\def\csname multinotes@#1cmd\endcsname##1{%
81     \ifundefined{switchcolumn}{}{\switchcolumn[\value{multinotes@#1number}]}%
82     \csname multinotes@#1style\endcsname%
83     \multinotes@selectlanguage{#1}%
84     ##1%
85     \multinotes@commonstyle\xspace%
86   }%
87   \renewenvironment{\csname multinotes@#1envname\endcsname}{%
88     \ifundefined{switchcolumn}{}{\switchcolumn[\value{multinotes@#1number}]}%
89     \csname multinotes@#1style\endcsname%
90     \multinotes@selectlanguage{#1}%
91   }{}%
92 }

```

```

93 \newcommand{\includelanguage}[1]{
94     \multinotes@activatelanguage{#1}
95     \LoadDictionaryFor{#1}{translations-basic-dictionary}
96     \LoadDictionaryFor{#1}{translations-theorem-dictionary}
97 }

```

6.2.3 Language style

Usage example: `\setlanguagestyle{welsh}{\slshape\color{red}}`

```

98 \newcommand{\setlanguagestyle}[2]{\expandafter\def\csname multinotes@#1style\endcsname{#2}}
99 \newcommand{\multinotes@commonstyle}{\normalfont\color{\multinotes@CurrentColour}}
100 \newcommand{\setcommonstyle}[1]{\renewcommand{\multinotes@commonstyle}{#1}}

```

6.2.4 Parallel typesetting

The *multicols* environment is based entirely on the *paracol* environment from the package of the same name. The number of columns required is obtained from the *numlangs* counter value. Any option commands for *paracol* such as `\columnsep`, `\columnseprule` or `\localcounter` can be used directly.

```

101 \newenvironment{multicols}{}{}
102 \newcommand{\multicolsync}{}
103 \newcommand{\multicolson}{
104     \ifnum\value{multinotes@numlangs}>1
105         \renewenvironment{multicols}{\begin{paracol}{\themultinotes@numlangs}}{\end{paracol}}
106         \renewcommand{\multicolsync}{\switchcolumn[0]*}
107     \fi
108 }
109 \newcommand{\multicolsoff}{
110     \renewenvironment{multicols}{}{}
111     \renewcommand{\multicolsync}{}
112 }
113 \AtBeginDocument{\multicolson}

```

Mark content to span across columns (used inside *multicols* environments)

```

114 \newcommand{\multicolspan}[1]{\multinotes@commonstyle%
115     \@ifundefined{switchcolumn}{}{%
116         \ifthenelse{\isempty{#1}}{%
117             \switchcolumn[0]%
118         }{%
119             \switchcolumn*[{#1}]%
120         }%
121     }%
122 }

```

6.2.5 Declarations

Finally some dubious TeX magic for dealing with declarations: when a language is excluded, everything between a corresponding language declaration and the

next non-alphabetical/numerical/space character should be removed. This doesn't always work as intended!

```

123 \def\multinotes@remove{\afterassignment\checkit\let\char=}%
124 \def\checkit{%
125     \ifcat\char\null %
126         \let\char=\null%
127     \fi%
128     \ifcat\char\space%
129         \let\next=\multinotes@remove%
130     \else%
131         \ifcat\char x%
132             \let\next=\multinotes@remove%
133         \else%
134             \ifcat\char 1%
135                 \let\next=\multinotes@remove%
136             \else%
137                 \noexpand\char
138                 \let\next=\relax%
139             \fi%
140         \fi%
141     \fi%
142     \next%
143 }%

```

6.3 Lecture notes

Macros for proof, answer and sketch names.

```

144 \def\multinotes@proofname{\GetTranslation{Proof}}
145 \def\multinotes@answername{\GetTranslation{Answer}}
146 \def\multinotes@sketchname{\GetTranslation{Sketch}}

```

Global variables to allow local override of the blanks, blankproofs, noproofs, blankanswers and noanswers options.

```

147 \newcommand{\blanksoff}{\multinotes@blanksfalse}
148 \newcommand{\blankson}{\multinotes@blankstrue}
149 \newcommand{\blankproofsoff}{\multinotes@blankproofsfalse}
150 \newcommand{\blankproofson}{\multinotes@blankproofstrue}
151 \newcommand{\blankanswersoff}{\multinotes@blankanswersfalse}
152 \newcommand{\blankanswerson}{\multinotes@blankanswerstrue}

153 \newcommand{\noproofsoff}{
154     \multinotes@noproofsfalse
155     \multinotes@createProofEnvironment
156 }
157 \newcommand{\noproofson}{
158     \multinotes@noproofstrue
159     \excludecomment{proof}
160 }
161 \newcommand{\noanswersoff}{
162     \multinotes@noanswersfalse
163     \multinotes@createAnswerEnvironment
164 }

```

```

165 \newcommand{\noanswerson}{
166     \multinotes@noanswerstrue
167     \excludecomment{answer}
168 }

```

Runtime flags

```

169 \newif\ifmultinotes@blankmode
170 \newif\ifmultinotes@imgblankmode

```

Colours

```

171 \newcommand{\multinotes@BackgroundColour}{white}
172 \newcommand{\multinotes@TextColour}{black}
173 \newcommand{\multinotes@ShowColour}{black}
174 \newcommand{\multinotes@ProofColour}{black}
175 \newcommand{\multinotes@SolutionColour}{black}
176 \newcommand{\multinotes@AnswerColour}{black}
177 \newcommand{\multinotes@QEDSymbol}{}
178 \AtBeginDocument{
179     \color{\multinotes@TextColour}
180     \pagecolor{\multinotes@BackgroundColour}
181 }

```

Set colours

```

182 \newcommand{\textcolour}[1]{\renewcommand{\multinotes@TextColour}{#1}}
183 \newcommand{\backgroundcolour}[1]{\renewcommand{\multinotes@BackgroundColour}{#1}}
184 \newcommand{\showcolour}[1]{\renewcommand{\multinotes@ShowColour}{#1}}
185 \newcommand{\proofcolour}[1]{\renewcommand{\multinotes@ProofColour}{#1}}
186 \newcommand{\answercolour}[1]{\renewcommand{\multinotes@AnswerColour}{#1}}
187 \newcommand{\includeqedsymbol}[1]{\renewcommand{\multinotes@QEDSymbol}{#1}}

```

Stretch factors for blank boxes (to accomodate handwriting)

```

188 \newcommand{\multinotes@StretchFactor}{1}
189 \newcommand{\multinotes@ImageStretchFactor}{1}
190 \newcommand{\setstretchfactor}[1]{\renewcommand{\multinotes@StretchFactor}{#1}}
191 \newcommand{\setimagestretchfactor}[1]{\renewcommand{\multinotes@ImageStretchFactor}{#1}}

```

Frame options

```

192 \newif\ifmultinotes@framedblankboxes
193 \newif\ifmultinotes@framedproofs
194 \newif\ifmultinotes@framedanswers
195 \newcommand{\framedblankboxes}{\multinotes@framedblankboxestruer}
196 \newcommand{\framedblankboxesoff}{\multinotes@framedblankboxesfalse}
197 \newcommand{\framedproofs}{\multinotes@framedproofstruer}
198 \newcommand{\framedproofsoff}{\multinotes@framedproofsfalse}
199 \newcommand{\framedanswers}{\multinotes@framedanswerstruer}
200 \newcommand{\framedanswersoff}{\multinotes@framedanswersfalse}
201 \newcommand{\setboxrule}[1]{\renewcommand{\multinotes@BoxRule}{#1}}

```

Runtime parameters

```

202 \newcommand{\multinotes@CurrentColour}{black}
203 \newcommand{\multinotes@CurrentBoxRule}{Opt}
204 \newcommand{\multinotes@CurrentBoxTitle}{}

```

```

205 \newcommand{\multinotes@BoxRule}{0pt}
206 \newcommand{\multinotes@makeboxtitle}{%
207     \textcolor{\multinotes@TextColour}{\noindent\multinotes@CurrentBoxTitle}%
208 }%

```

6.4 Inline boxes

Inline blankbox

```

209 \newcommand\multinotes@inlineboxsize{normal}
210 \newcommand{\inlineboxsize}[1]{\renewcommand{\multinotes@inlineboxsize}{#1}}
211 \newcommand{\blank}[1]{#1}
212 \ifmultinotes@blanks
213     \renewtcbox{\blank}{
214         on line,
215         size=\multinotes@inlineboxsize,
216         colback=\multinotes@BackgroundColour,
217         coltext={\ifmultinotes@blanks\multinotes@BackgroundColour\else\multinotes@ShowColour\fi}
218         boxrule={\ifmultinotes@blanks\multinotes@BoxRule\elseOpt\fi}
219     }
220 \fi

```

Inline fillbox

```

221 \newcommand{\fillbox}[1]{#1}
222 \ifmultinotes@blankanswers\else
223     \renewtcbox{\fillbox}{
224         on line,
225         size=\multinotes@inlineboxsize,
226         arc=0pt,
227         boxsep=1pt,left=10pt,right=10pt,
228         boxrule=\multinotes@BoxRule,
229         colback=\multinotes@BackgroundColour,
230         coltext={
231             \ifmultinotes@noanswers\multinotes@BackgroundColour
232             \else\ifmultinotes@blankanswers\multinotes@BackgroundColour
233             \else\multinotes@AnswerColour
234             \fi\fi
235         },
236     }
237 \fi

```

Inline answer box

```

238 \newcommand{\ans}[1]{#1}
239 \ifmultinotes@noanswers\else
240     \ifthenelse{\isempty{#1}}{
241         {\renewcommand{\multinotes@CurrentBoxTitle}{#1}}%
242         {\renewcommand{\multinotes@CurrentBoxTitle}{}}%
243     }{\renewtcbox{\ans}{
244         on line,
245         title=\sl\multinotes@CurrentBoxTitle,
246         coltitle=\multinotes@AnswerColour,
247         attach title to upper={\strut},
248         size=\multinotes@inlineboxsize,

```

```

249     arc=1pt,
250     boxsep=1pt,left=5pt,right=5pt,
251     boxrule=\multinotes@BoxRule,
252     colback=\multinotes@BackgroundColour,
253     coltext={
254         \ifmultinotes@blankanswers\multinotes@BackgroundColour
255         \else\ifmultinotes@noanswers\multinotes@BackgroundColour
256         \else\multinotes@AnswerColour
257         \fi\fi
258     },
259 }
260 \fi

```

Inline proof box

```

261 \newcommand{\prf}[1]{
262 \ifmultinotes@noproofs\else
263     \ifthenelse{\isempty{#1}}{
264         {\renewcommand{\multinotes@CurrentBoxTitle}{#1}}%
265         {\renewcommand{\multinotes@CurrentBoxTitle}{}}%
266     \renewtcbox{\prf}{
267         on line,
268         title=\sl\multinotes@CurrentBoxTitle,
269         coltitle=\multinotes@ProofColour,
270         attach title to upper={\strut},
271         size=\multinotes@inlineboxsize,
272         arc=1pt,
273         boxsep=1pt,left=5pt,right=5pt,
274         boxrule=\multinotes@BoxRule,
275         colback=\multinotes@BackgroundColour,
276         coltext={
277             \ifmultinotes@blankproofs\multinotes@BackgroundColour
278             \else\ifmultinotes@noproofs\multinotes@BackgroundColour
279             \else\multinotes@ProofColour
280             \fi\fi
281         },
282     }
283 \fi

```

6.5 Display boxes

Box template

```

284 \newenvironment{multinotes@tcbox}{%
285     \begin{tcolorbox}[
286         breakable,
287         notitle,
288         boxrule={\multinotes@BoxRule},
289         colback={\multinotes@BackgroundColour},
290         before={\smallskip},
291         after={},
292         coltext={\multinotes@CurrentColour},
293         skin=enhanced jigsaw,
294         opacityback=0,

```

```

295         height fixed for = first and middle,
296         ignore nobreak,
297         before upper={\parindent0em\noindent},
298         flushleft upper
299     ]%
300     \begingroup
301 }{
302     \endgroup
303     \end{tcolorbox}
304 }

```

Basic display box

```

305 \newenvironment{multinotes@basicbox}[1][{}]{
306     \ifthenelse{\isempty{#1}}{
307         {\renewcommand{\multinotes@CurrentBoxTitle}{}}{
308             {\renewcommand{\multinotes@CurrentBoxTitle}{#1}}
309         \ifmultinotes@blankmode
310             \renewcommand{\multinotes@CurrentColour}{\multinotes@BackgroundColour}
311             \setstretch{\multinotes@StretchFactor}
312             \begin{multinotes@tcbox}%
313             \ifthenelse{\isempty{#1}}{ }\{\multinotes@makeboxtitle\}\quad
314         \else
315             \par
316             \ifthenelse{\isempty{#1}}{ }\{\multinotes@makeboxtitle\}\quad
317             \color{\multinotes@CurrentColour}
318         \fi
319     }{
320         \ifmultinotes@blankmode
321             \end{multinotes@tcbox}
322             \setstretch{1}
323         \fi
324         \multinotes@blankmodefalse
325         \color{\multinotes@TextColour}
326     }

```

Basic framed box

```

327 \newenvironment{multinotes@framedbox}[1][{}]{
328     \ifthenelse{\isempty{#1}}{
329         {\renewcommand{\multinotes@CurrentBoxTitle}{}}{
330             {\renewcommand{\multinotes@CurrentBoxTitle}{#1}}
331         \ifmultinotes@blankmode
332             \renewcommand{\multinotes@CurrentColour}{\multinotes@BackgroundColour}
333             \setstretch{\multinotes@StretchFactor}
334         \else
335             \color{\multinotes@CurrentColour}
336         \fi
337         \begin{multinotes@tcbox}%
338         \ifthenelse{\isempty{#1}}{ }\{\multinotes@makeboxtitle\}
339     }{
340     \mbox{}
341     \end{multinotes@tcbox}
342     \ifmultinotes@blankmode
343         \setstretch{1}

```

```

344 \fi
345 \multinotes@blankmodefalse
346 \renewcommand{\multinotes@CurrentColour}{\multinotes@TextColour}
347 \color{\multinotes@TextColour}
348 }

```

Blankbox

```

349 \newenvironment{blankbox}[1][]{
350   \def\boxtitle{}
351   \ifthenelse{\isempty{#1}}{\def\boxtitle{#1}}
352   \renewcommand{\multinotes@CurrentColour}{\multinotes@ShowColour}
353   \ifmultinotes@blanks
354     \multinotes@blankmodetrue
355   \fi
356   \ifmultinotes@framedblankboxes
357     \begin{multinotes@framedbox}[\sl\boxtitle]
358   \else
359     \begin{multinotes@basicbox}[\sl\boxtitle]
360   \fi
361 }{%
362   \ifmultinotes@framedblankboxes
363     \end{multinotes@framedbox}
364   \else
365     \end{multinotes@basicbox}
366   \fi
367 }

```

Proof box

```

368 \newcommand\multinotes@createProofEnvironment{
369   \let\proof\@undefined
370   \let\endproof\@undefined
371   \newenvironment{proof}[1][]{
372     \renewcommand{\multinotes@CurrentColour}{\multinotes@ProofColour}
373     \ifmultinotes@blankproofs
374       \multinotes@blankmodetrue
375     \fi
376     \ifthenelse{\isempty{##1}}{
377       \ifmultinotes@framedproofs
378         \begin{multinotes@framedbox}[\sl\multinotes@proofname]
379       \else
380         \begin{multinotes@basicbox}[\sl\multinotes@proofname]
381       \fi
382     }{%
383       \ifmultinotes@framedproofs
384         \begin{multinotes@framedbox}[##1]
385       \else
386         \begin{multinotes@basicbox}[##1]
387       \fi
388     }
389   }{%
390     \mbox{} \hfill \multinotes@QEDSymbol
391     \ifmultinotes@framedproofs
392       \end{multinotes@framedbox}

```

```

393         \else
394             \end{multinotes@basicbox}
395         \fi
396     }
397 }
398 \ifmultinotes@noproofs
399     \excludecomment{proof}
400 \else
401 \multinotes@createProofEnvironment
402 \fi

```

`answer` (*env.*) *Answer box*

```

403 \newcommand\multinotes@createAnswerEnvironment{
404     \let\answer\@undefined
405     \let\endanswer\@undefined
406     \newenvironment{answer}[1][\]{
407         \renewcommand{\multinotes@CurrentColour}{\multinotes@AnswerColour}
408         \ifmultinotes@blankanswers
409             \multinotes@blankmodetrue
410         \fi
411         \ifthenelse{\isempty{##1}}{
412             \ifmultinotes@framedanswers
413                 \begin{multinotes@framedbox}[\sl\multinotes@answername]%
414             \else
415                 \begin{multinotes@basicbox}[\sl\multinotes@answername]%
416             \fi
417         }{%
418             \ifmultinotes@framedanswers
419                 \begin{multinotes@framedbox}[##1]%
420             \else
421                 \begin{multinotes@basicbox}[##1]%
422             \fi
423         }
424     }{
425         \mbox{} \hfill \multinotes@QEDSymbol
426         \ifmultinotes@framedanswers
427             \end{multinotes@framedbox}
428         \else
429             \end{multinotes@basicbox}
430         \fi
431     }
432 }
433 \ifmultinotes@noanswers
434     \excludecomment{answer}
435 \else
436 \multinotes@createAnswerEnvironment
437 \fi

```

6.6 Floats for sketches, figures and videos

`\sketchbox` *Custom float for sketch boxes. Caption `\sketchname` is always typeset at the top-left. Mandatory argument is the height of the box as a proportion of `\textheight`.*

Optional argument is a subtitle which appears in parentheses after the caption.

```

438 \floatstyle{boxed}
439 \newfloat{sketch}{!ht}{losk}
440 \newcommand{\sketchbox}[2][{}]{
441 \ifmultinotes@nosketchboxes\else
442   \renewcommand{\sketchbox}[2][{}]{%
443     \begin{sketch}[H]%
444       \color{\multinotes@CurrentColour}
445       \vspace{1ex}\hspace{1ex}\textsl{\multinotes@sketchname}%
446       \ifthenelse{\isempty{#1}}{#{#1}}{\par%
447         \parbox{\textwidth}{\vspace{#2\textheight}}}%
448       \end{sketch}%
449     }
450 \fi

```

`\includevideo` *Custom float for videos*

```

451 \newcommand{\includevideo}[2][1]{\url{#2}}
452 \floatstyle{boxed}
453 \newfloat{video}{!ht}{lov}[section]
454 \floatname{video}{Video}
455 \captionsetup[video]{labelfont=normalfont}

```

Make \includegraphics respond to blankmode

```

456 \let\oldincludegraphics=\includegraphics
457 \renewcommand\includegraphics[2][{}]{
458   \ifmultinotes@blankmode
459     \scalebox{\multinotes@ImageStretchFactor}{
460       \phantom{\oldincludegraphics[#1]{#2}}
461     }
462   \else
463     \oldincludegraphics[#1]{#2}
464   \fi
465 }

```

Display alternative image under under blankmode

```

466 \newlength\imageheight
467 \newlength\imagewidth
468 \newcommand{\includetwographics}[3][scale=1]{
469   \ifmultinotes@blankmode
470     \settoheight{\imageheight}{\oldincludegraphics[#1]{#2}}
471     \settowidth{\imagewidth}{\oldincludegraphics[#1]{#2}}
472     \scalebox{\multinotes@ImageStretchFactor}{
473       \oldincludegraphics[height=\imageheight,width=\imagewidth,keepaspectratio]{#3}
474     }
475   \else
476     \oldincludegraphics[#1]{#2}
477   \fi
478 }

```

Replace tikz picture by blank box (moody)

```

479 \newcommand{\blanktikz}{%

```

```

480 \ifmultinotes@blankmode
481 \tikzset{every picture/.append style={
482     scale=\multinotes@ImageStretchFactor,
483     execute at end picture={
484         \draw[fill=\multinotes@BackgroundColour]
485             (current bounding box.south west)
486             rectangle
487             (current bounding box.north east);
488     }}}
489 \fi
490 }

```

6.7 Exercise sheets (inspired by exam.cls)

Check that exam.cls is not already loaded.

```
491 \@ifclassloaded{exam}{}{%
```

Counters and labels

```

492 \newcounter{question}
493 \newcounter{partno}
494 \newcounter{subpart}
495 \newcounter{choice}
496 \newcommand\questionlabel{\arabic{question}.}
497 \newcommand\partlabel{(\alph{partno})}
498 \newcommand\subpartlabel{(\roman{subpart})}

```

Default symbols (circle for MC, square for MA)

```

499 \newcommand\choicechar[1]{\def\choice@char{#1}}
500 \newcommand\chosenchar[1]{\def\chosen@char{#1}}
501 \newcommand\checkboxchar[1]{\def\checkbox@char{#1}}
502 \newcommand\checkedchar[1]{\def\checked@char{#1}}
503 \newcommand\bigsquare{\raisebox{0.5ex}{\fbox{\phantom{\rule{0.5ex}{0.5ex}}}}}
504 \choicechar{$\bigcirc$}
505 \chosenchar{$\text{\rlap{\,$\checkmark$}}\bigcirc$}
506 \checkboxchar{$\bigsquare$}
507 \checkedchar{$\text{\rlap{\hspace{0.2ex}\raisebox{0.2ex}{\checkmark}}}\bigsquare$}

```

Hooks for customisation

```

508 \newcommand\questionshook{}
509 \newcommand\partshook{}
510 \newcommand\subpartshook{}
511 \newcommand\choiceshook{}
512 \newcommand\checkboxeshook{}

```

Questions, parts and subparts.

```

513 \newenvironment{questions}{
514     \def\@queslevel{question}
515     \def\question{\@checkqueslevel{question}\item}
516     \def\subpart{\@checkqueslevel{subpart}\item}
517     \list{\questionlabel}{
518         \usecounter{question}

```

```

519         \settowidth{\leftmargin}{10.\hskip\labelsep}
520         \labelwidth\leftmargin\advance\labelwidth-\labelsep
521         \partopsep=0pt\questionshook
522     }}{\endlist}
523 \newenvironment{parts}{
524     \def\@queslevel{part}
525     \def\part{\@checkqueslevel{part}\item}
526     \list{\partlabel}{
527         \usecounter{partno}\def\makelabel##1{\hss\llap{##1}}
528         \settowidth{\leftmargin}{(m)\hskip\labelsep}
529         \labelwidth\leftmargin\advance\labelwidth-\labelsep
530         \topsep=0pt\partopsep=0pt\partshook
531     }}{\endlist}
532 \newenvironment{subparts}{
533     \def\@queslevel{subpart}
534     \list{\subpartlabel}{
535         \usecounter{subpart}\def\makelabel##1{\hss\llap{##1}}
536         \settowidth{\leftmargin}{vii.\hskip\labelsep}
537         \labelwidth\leftmargin\advance\labelwidth-\labelsep
538         \topsep=0pt\partopsep=0pt\subpartshook
539     }}{\endlist}

```

Choices and checkboxes (MC and MA)

```

540 \newif\if@correctchoice
541 \newcommand\CorrectChoiceEmphasis[1]{\def\CorrectChoice@Emphasis{#1}}
542 \CorrectChoiceEmphasis{\bfseries}
543 \newenvironment{choices}{
544     \list{\choice@char}{
545         \usecounter{choice}
546         \settowidth{\leftmargin}{W.\hskip\labelsep}
547         \def\choice{\if@correctchoice\endgroup\fi\item}
548         \def\CorrectChoice{
549             \if@correctchoice\endgroup\fi
550             \ifmultinotes@noanswers\item
551             \else\ifmultinotes@blankanswers\item
552             \else
553                 \begingroup
554                 \@correctchoicetrue
555                 \CorrectChoice@Emphasis
556                 \item[\chosen@char]
557             \fi\fi
558         }
559         \let\correctchoice\CorrectChoice
560         \topsep=2pt\partopsep=0pt\choiceshook
561     }
562 }{\if@correctchoice\endgroup\fi\endlist}
563 \newenvironment{checkboxes}{
564     \list{\checkbox@char}{
565         \usecounter{choice}
566         \settowidth{\leftmargin}{W.\hskip\labelsep}
567         \def\choice{\if@correctchoice\endgroup\fi\item}
568         \def\CorrectChoice{
569             \if@correctchoice\endgroup\fi
570             \ifmultinotes@noanswers\item

```

```

571             \else\ifmultinotes@blankanswers\item
572             \else
573                 \begingroup
574                 \@correctchoicetrue
575                 \CorrectChoice@Emphasis
576                 \item[\checked@char]
577                 \fi\fi
578             }
579             \let\correctchoice\CorrectChoice
580             \topsep=2pt\partopsep=0pt\choiceshook
581         }
582     }\if@correctchoice\endgroup\fi\endlist}

```

Nesting check and indentation trick.

```

583     \def\@checkqueslevel#1{
584         \begingroup
585         \def\@temp{#1}
586         \ifx\@temp\@queslevel\null
587         \else\ClassError{multinotes}{
588             I found a #1 where I expected to find a \@queslevel
589             \MessageBreak
590         }{}
591         \fi\endgroup
592     }
593     \long\def\uplevel#1{
594         \par\bigskip\vbox{
595             \leftskip=\@totalleftmargin
596             \advance\leftskip-\leftmargin
597             \advance\@totalleftmargin-\leftmargin
598             \advance\linewidth\leftmargin
599             #1
600         }
601         \nobreak
602     }

```

End \@ifclassloaded{exam} check.

```

603 }

```